

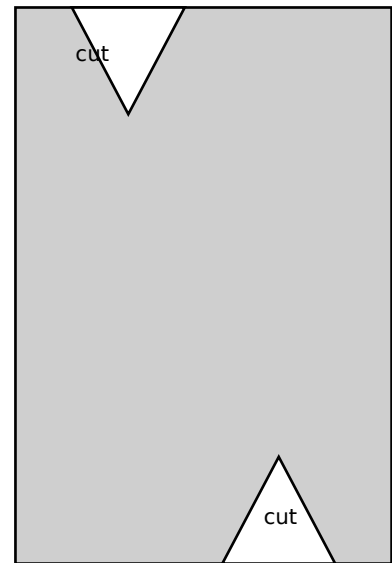
Set 3 - Composite Figures

Level: Medium to Hard

Name: _____ Class: _____ Date: _____

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1. A rectangle measures 36 cm by 20 cm. Two unshaded triangles are cut away. Each triangle has base 12 cm and height 8 cm. Find the shaded area.

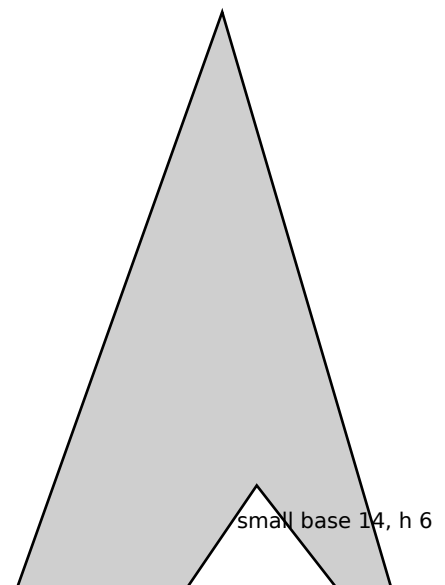


rectangle 36 x 20; each cut 12 x 8

Working:

Answer:

2. In Triangle ABC, BC = 42 cm and the height from A to BC is 18 cm. If the unshaded triangle has base 14 cm and height 6 cm, find the shaded area.



large triangle: base 42, h 18

Working:

Answer: _____

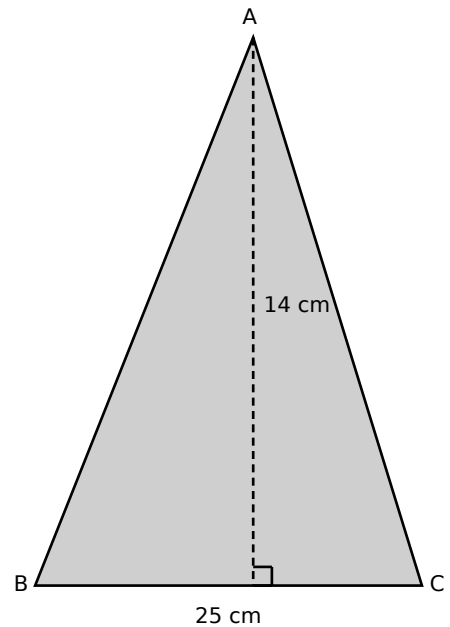
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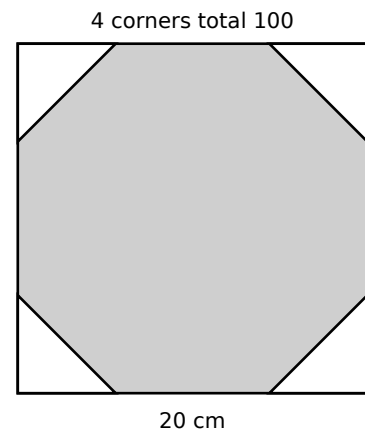
3. A triangle has base 25 cm and height 14 cm. Find its area.



Working:

Answer:

4. A square has side 20 cm. Four unshaded corner triangles have a total area of 100 cm^2 . Find the shaded area.



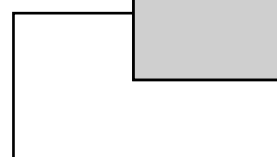
Working:

Answer:

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5. Two identical right-angled triangles each have area 60 cm^2 . They overlap by 18 cm^2 . Find the area covered by the two triangles together.

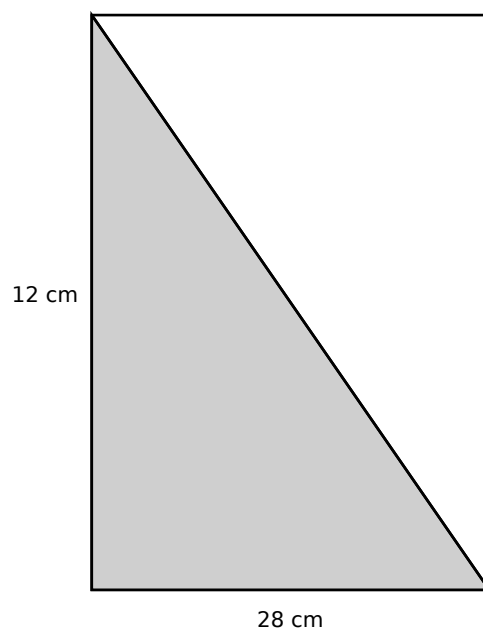


2 shapes, overlap 18

Working:

Answer:

6. In Rectangle ABCD, $AB = 28 \text{ cm}$ and $AD = 12 \text{ cm}$. Point E is on DC such that $EC = 18 \text{ cm}$. Find the area of shaded Triangle AEC.



Working:

Answer: _____

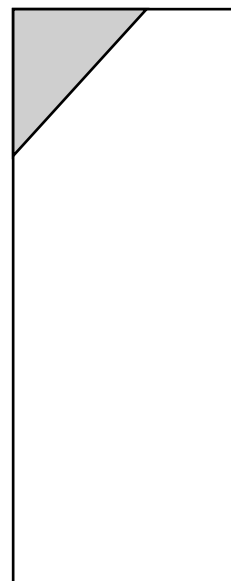
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7. A rectangle measuring 24 cm by 10 cm has a triangular corner folded over. The folded triangle has base 10 cm and height 8 cm. Find the remaining visible area.

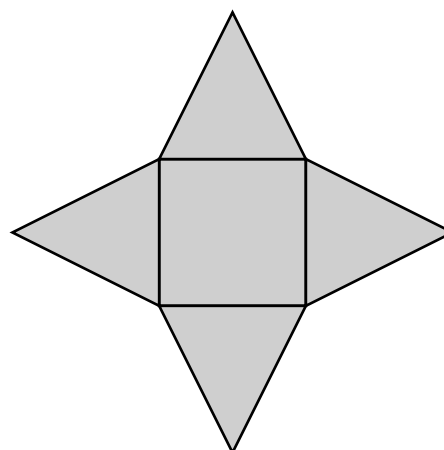


rect 24 x 10; fold 10 x 8

Working:

Answer:

8. A figure is made up of a square of area 64 cm^2 and four identical right-angled triangles. Each triangle has base 8 cm and height 6 cm. Find the total area.



square area 64 + 4 triangles

Working:

Answer: _____

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- 9.** Rina had some marbles. She gave $\frac{1}{6}$ of them to a friend and $\frac{2}{5}$ of the remainder to her cousin. She had 40 marbles left. How many marbles did she have at first?

Working:

Answer:

- 10.** Ali spent $\frac{3}{7}$ of his money on books and $\frac{1}{4}$ of the remainder on files. He had \$90 left. How much money did he have at first?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	624 cm ²	Rectangle = 720; two triangles = $2 \times 48 = 96$; shaded = 624 cm ²
2	336 cm ²	Large = $\frac{1}{2} \times 42 \times 18 = 378$; small = $\frac{1}{2} \times 14 \times 6 = 42$; shaded = 336 cm ²
3	175 cm ²	$\frac{1}{2} \times 25 \times 14 = 175$ cm ²
4	300 cm ²	Square = $20 \times 20 = 400$; shaded = $400 - 100 = 300$ cm ²
5	102 cm ²	Combined area = $60 + 60 - 18 = 102$ cm ²
6	108 cm ²	Base EC = 18, height = 12. Area = $\frac{1}{2} \times 18 \times 12 = 108$ cm ²
7	200 cm ²	Rectangle = 240; folded triangle = 40; remaining = 200 cm ²
8	160 cm ²	Each triangle = 24; 4 triangles = 96; total = $64 + 96 = 160$ cm ²
9	80 marbles	Left = $\frac{5}{6} \times \frac{3}{5} = \frac{1}{2}$ of total. Total = 80
10	\$210	Left = $\frac{4}{7} \times \frac{3}{4} = \frac{3}{7}$ of total. $\frac{3}{7} = 90$, so total = 210